

Extraction of the midsagittal plane from morphological neuroimages using the Kullback–Leibler’s measure

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Abstract

A theoretically simple and computationally efficient method to extract the midsagittal plane (MSP) from volumetric neuroimages is presented. The method works in two stages (coarse and fine) and is based on calculation of the Kullback and Leibler’s (KL) measure, which characterizes the difference between two distributions. Slices along the sagittal direction are analyzed with respect to a reference slice to determine the coarse MSP. To calculate the final MSP, a local search algorithm is applied. The proposed method does not need any preprocessing, like reformatting, skull stripping, etc. The algorithm was validated quantitatively on 75 MRI datasets of different pulse sequences (T1WI, T2WI, FLAIR and SPGR) and MRA. The angular and distance errors between the calculated MSP and the ground truth lines marked by the expert were calculated. The average distance and angular deviation were 1.25 pixels and 0.63°, respectively. In addition, the algorithm was tested qualitatively on PD, FLAIR, MRA, and CT datasets. To analyze the robustness of the method against rotation, inhomogeneity and noise, the phantom data were used.

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1. Introduction

Though the human brain exhibits a common gross anatomy, there exist substantial individual variations in shape, size and position of cerebral structures. Moreover, brain scans vary due to variability in anatomy and imaging conditions. To perform brain image comparison, the scans have to be first registered with respect to a reference brain image. One of the key landmarks in this registration process is the interhemispheric fissure, which is generally approximated by the midsagittal plane (MSP) (Talairach and Tournoux, 1988). The knowledge of MSP not only reduces the number of degrees of freedom in the registra-

tion process, but also enables comparing both hemispheres and facilitates the calculation of landmarks located on the MSP, such as the anterior (AC) and posterior (PC) commissures. The AC and PC landmarks are important as major brain atlases, the Talairach–Tournoux atlas (Talairach and Tournoux, 1988) and the Schaltenbrand–Wahren atlas (Schaltenbrand and Wahren, 1977), are based on them. The MSP can also be used for detection of bilateral asymmetries caused by mass effects and lesions (Liu et al., 2001; Joshi et al., 2003; Wang et al., 2004), and to assess change in brain symmetry in pathological processes, such as schizophrenia (Crow, 1990).

Reviews of MSP extraction methods can be found in Liu et al. (2001), Prima et al. (2002), Hu and Nowinski (2003). Some of the methods are based on interhemispheric fissure identification and symmetry criteria (Marais et al., 1996). They exploit the extraction of symmetry lines in axial or coronal 2D slices and fitting a 3D plane to the dataset (Hu and Nowinski, 2003; Guillemaud et al., 1996). There

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are methods based on cross-correlation and symmetry measures (Liu et al., 2001; Ardekani et al., 1997; Tuzikov et al., 2003), but they are sensitive to asymmetry (Prima et al., 2002; Junck et al., 1990). A large deformation image warping method (Joshi et al., 2003) is used for finding the plane of symmetry, not necessarily being the MSP.

We briefly summarized features for several methods in Table 1.

Problem of symmetry detection arises in different areas, e.g. computer vision, object recognition, shape representation, etc. Different approaches for this problem can be found in Gofman and Kiryati (1996), Sun (1995), Sun and Sherrah (1997).

The Kullback–Leibler's (KL) measure can be used to measure the difference between two distributions. It has been extensively used in registration of image datasets (Chan et al., 2003; Plum et al., 2003) and in segmentation (Hibbard, 2004). We propose the KL-measure for construction of the MSP directly from a 3D volume. To our best knowledge no work on Kullback–Leibler's measure based identification of the MSP has been reported yet. Our results show that the algorithm is applicable for different modalities and is able to handle pathological cases. The

algorithm can also provide a good initial estimate for other algorithms, for example Joshi et al. (2003), where the estimation of the plane of symmetry is based on iterative optimization algorithm in the space of affine motions $GL(3) \times R^3 \subset R^{12}$.

The closest related work to our method is Smith and Jenkinson (1999). It calculates the symmetry score for all possible mid-plane orientations, i.e., for all angles α and β . For each mid-plane given by these angles the cumulative symmetry score for all lines perpendicular to this mid-plane is calculated and center of symmetry is found. The center of symmetry with a maximum score defines the 3D symmetry plane. The method is time consuming, taking 40 min to complete calculations (Table 1). The faster method proposed by Hu and Nowinski (2003) is based on local symmetry and outlier removal calculating the MSP in less than 6 s. This method works across a stack of a few 2D images and exploits contrast between white matter (WM), grey matter (GM), and cerebrospinal fluid (CSF) in the vicinity of the interhemispheric fissure (IF). The algorithm extracts the MSP in four steps: (1) determines suitable axial slices for processing (16 slices are selected), (2) localizes the fissure line segments on them, (3) selects inliers from the

Table 1
Brief summary of the features of some MSP extraction methods

Reference	Principle	Modality	Accuracy	Performance
Smith and Jenkinson (1999)	Calculation of the measure based on the global symmetry of the lines orthogonal to the candidate plane	MR (T1WI, T2WI, EPI), PET, SPECT, and CT	Quantitative value not provided	Forty minutes for typical MRI case (Silicon Graphics Origin 200)
Hu and Nowinski, 2003	Method is based on a local anatomical symmetry in the neighborhood of the interhemispheric fissure (IF) and outliers removal. It samples the IF region with gray matter, white matter and cerebrospinal fluid by means of five pairs of parallel lines searching for the maximum symmetry	MR (T1WI, T2WI, FLAIR, SPGR), CT	Average angular error of 0.43° and the average distance error of 0.65 mm for an isotropic reformatted volume	Less then 6 s (Pentium 4, 2.4 GHz CPU, 1G RAM)
Prima et al. (2002)	Computation of local similarity measures between the two sides of the head using a block matching procedure	MR (T1WI, T2WI, PD), CT, PET, SPECT	The qualitative visual evaluation of the patient cases has been carried out by clinicians	3 min (PC, OS Linux, 450 MHz, 256M RAM)
Brummer (1991)	Detection of the longitudinal fissure in 3D based on edge detection and the Hough transform principle	MR (T1WI, T2WI)	The results were verified visually by displaying the intersection line of the detected IF plane overlaid on the slice images	20 s (SUN/Pixar Image computer) for 40-slice data set
Liu et al. (2001)	Estimation of the mid-sagittal plane from an edge image	MR (T1WI, T2WI), CT	The detection accuracy of yaw and roll angles on the resampled images are estimated within 1° of rotated angle values. Method is robust to rotation	About 7 min (SGI O2 R10000) for a 20-slice neural image

extracted fissure line segments through histogram-based outlier removal, and (4) calculates the equation of the MSP from the selected inliers by means of the least square error fit. The fissure line segments are localized by minimizing the local symmetry index characterizing anatomical properties of images in the vicinity of the IF.

Our method extracts the MSP directly in 3D and is equally fast (extraction in average takes about 5 s). In addition it works successfully for CT and MRI scans, including various pulse sequences (T1, T2, PD, FLAIR, SPGR, MRA), low contrast images, and pathological cases.

2. Method

2.1. Definitions

The concept of unexpectedness or surprise according to Renyi (Renyi, 1961) can be explained as follows. Consider a random discrete variable X with probability distribution $p = \{p_i\}$, where p_i is the probability for the system to be in i th state. The quantity $\log(1/p_i)$ is called the unexpectedness or surprise (Renyi, 1961). If $p_i = 1$, then the event is certain to happen and no surprise is expected. If the event is nearly impossible ($p_i \approx 0$), it means an infinite surprise when it occurs. Consider two discrete distributions $p = \{p_i\}$ and $q = \{q_i\}$, where p_i and q_i are the probabilities of encountering of the i th state in a set of system states. The change of unexpectedness of the probability p with respect to the probability q can be defined as $\log(1/q_i) - \log(1/p_i)$. By averaging the unexpectedness of the events over p_i , we get

$$I(p/q) = \sum_i p_i \log p_i - \sum_i p_i \log q_i = \sum_i p_i \log(p_i/q_i) \quad (1)$$

Function $I(p/q)$ is known as the Kullback and Leibler's measure (KL-measure) (Kullback and Leibler, 1951) of information associated with two probability distributions p and q . This is a measure of difference between two distributions. Eq. (1) can also be rewritten as

$$I(p/q) = H(p) - K(p, q) \quad (2)$$

where $H(p) = -\sum_i p_i \log p_i$ is the negative of Shannon's entropy (Shannon, 1948) and $K(p, q) = \sum_i p_i \log q_i$ is the measure of information generally referred as Kerridge's inaccuracy (Kerridge, 1961).

2.2. Problem statement

Let $p = \{p_i\}$ and $q = \{q_i\}$ be the discrete probability distributions of intensities for two sagittal slices. The values of p_i and q_i denote the probability of occurrence of pixels with intensity i in the 2D slice and reference slice, respectively, and are calculated from the histograms. Using the KL-measure we can define the change in the probability of the intensity in two slices. We use the following assumptions (Kullback, 1959): (1) $0 \log(0/p_i) = 0$; (2) terms in Eq. (1) where the denominator is zero are undefined and

must be neglected to provide absolute continuity of p_i with q_i .

The midsagittal plane (MSP) can be defined (Talairach and Tournoux, 1988, p. 5) as a plane passing through the interhemispheric fissure (IF), about which structures (such as the AC, PC, ventricles, thalamus, corpus callosum) exhibit an approximate bilateral symmetry. In normal datasets, the MSP can be considered as a plane, which has the maximum amount of cerebrospinal fluid (CSF) amongst all sagittal slices. In an ideal case, where the IF is not curved and the sagittal size of voxel is smaller than the width of IF, the MSP typically contains the following cerebral structures:

- AC and PC,
- CSF of the third and fourth ventricles, aqueduct and that of the IF,
- cross-sections of structures including the corpus callosum, brainstem, pineal body, cerebellum, and massa intermedia (if it exists),
- no cross-sections of the cortical areas.

This anatomical content differentiates the MSP from the other planes and is clearly reflected in the intensity histogram. A typical histogram of the MSP of a T1 weighted volume shows a significant peak at CSF and a low peak at WM, as opposed to a histogram of a slice apart from the MSP with a low peak at CSF and a high peak at WM (Fig. 1).

As we consider the occurrence of intensity of pixels in the MSP as events, the probability of occurrence of CSF pixels is rather high (which means the entropy is low and

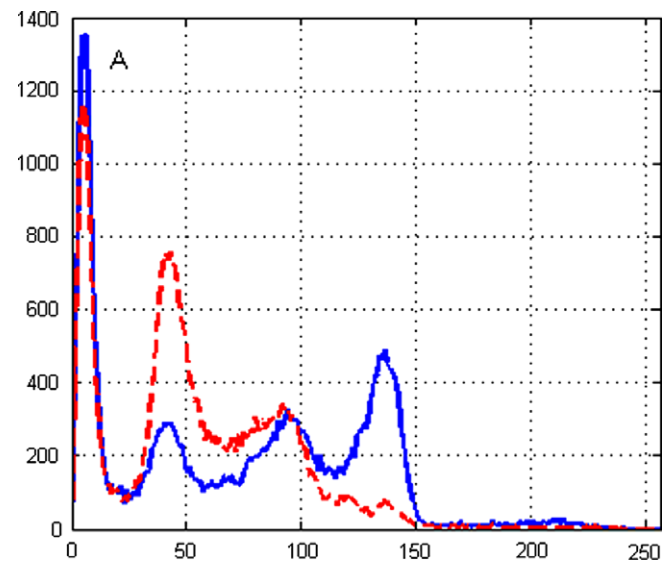


Fig. 1. Histograms for sagittal slices (the horizontal axis is intensity and vertical the number of pixels in the image) for BrainWeb MRI phantom image (3% noise, 20% inhomogeneity, volume's dimensions of $181 \times 217 \times 181$, 1 mm thickness). The plot for the MSP slice is shown in the dashed line and that for the slice 2 cm apart is represented as the solid line. Peak (A) corresponds to the background.

unexpectedness of this event is low because of a large portion of the MSP is CSF). Therefore, we can expect the MSP to have lower entropy than that of the neighboring slices.

Entropy is a slice specific measure, which is sensitive to noise and inhomogeneities. In our study the KL-measure is applied to capture the change in probabilities of gray values between the reference slice and the slice under testing. Both measures are dependent on the frequency of occurrence of intensity but not on the intensity. However, the KL-measure is a relative measure, and includes volume specific elements like noise, inhomogeneity, etc. of both slices (reference and testing) in the calculation, which acts as a form of normalization. The comparative plots of Shannon entropy measure and the KL-measure applied for a routine scan MRI volume dataset are shown in Fig. 2. There is no one-to-one correspondence between the minimum of entropy and the maximum of KL-measure in most cases (as these measures are different, see Eq. (2)). The difference between them is addressed in the Section 4.

There is another explanation from different viewpoint of why the MSP is close to the extremum of entropy measure and KL-measure. A non-constant continuous function of intensity histogram, defined on slices along the sagittal direction will have a roughly symmetrical plot about the MSP and the extremum point in the vicinity of the MSP, due to the fact that the brain is approximately symmetrical about the MSP. Generally the plot of this function is not symmetrical because the human brain is not symmetrical by itself (Toga and Thompson, 2003). Several works (e.g. Liu et al., 2001) have assumed that the brain has maximum symmetry about the MSP.

We evaluated several functions (entropy, energy, etc) and concluded that the Kullback–Leibler’s measure gave the best approximation of the MSP.

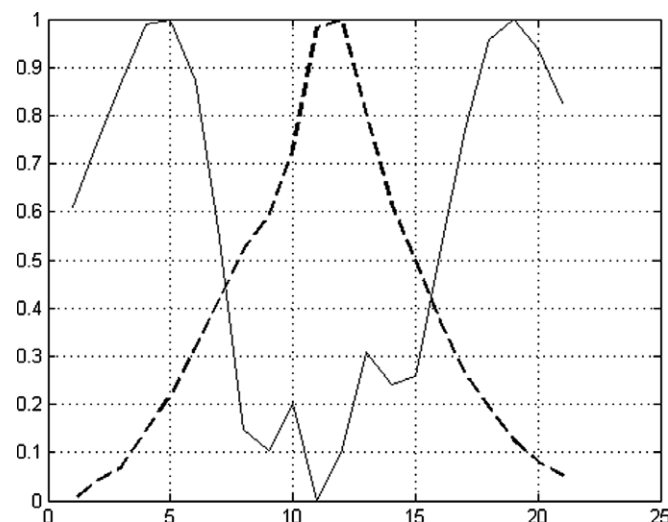


Fig. 2. Plots of normalized Shannon entropy (the solid line) and the KL-measure (the dashed line). The horizontal axis is the slice number in the volume.

2.3. Algorithm

Our algorithm works in two stages. Initially it calculates a coarse approximation of the MSP (cMSP) by analyzing the sagittal slices of a given brain volume data. In the second stage, the cMSP is refined to obtain the best approximation to the MSP.

Let (v_x, v_y, v_z) denote the voxel size in millimeters. The voxel coordinates are assumed to be integers. The X coordinate axis of the volume dataset corresponds to the right–left direction, Y to anterior–posterior direction, and Z to superior–inferior direction.

In the first stage, the following steps are carried out:

- Read the brain volume data, voxel size, and orientation. The initial volume can be either isotropic or anisotropic.
- Generate sagittal slices, if the original orientation is axial or coronal.
- Identify the central slice in the volume along the sagittal direction.
- Define a volume of interest (VOI) in the range of ± 2 cm around the central slice in the sagittal direction.
- Chose the first slice of the VOI as the reference slice.
- Compute the KL-measure for all the slices in the VOI with respect to the reference slice.
- Select the slice with the maximum KL-measure. This slice is referred as cMSP (Fig. 3).

Let s be the slice number at which the KL-measure attains its maximum value. The four corner points of the cMSP are (s, y_1, z_1) , (s, y_2, z_2) , (s, y_3, z_3) , (s, y_4, z_4) . In the second (fine processing) stage, the MSP is found using the coordinates of cMSP obtained in the first stage. The corner points of the cMSP are perturbed to obtain the plane giving the maximum KL-measure. The details of finding the final coordinates are as follows:

- Three corner points of cMSP are chosen and are represented by (s, y_i, z_i) , where $i = 1, 2, 3$. (Any three corner points may be used because they define the same plane.)
- Define the reference slice at the distance of 2 cm from the cMSP slice in the sagittal direction. This reference slice remains the same for the whole second stage.
- Let L be the number of sagittal slices in the VOI. A step size of $\Delta_0 = \min(\lceil L/8 \rceil, 8)$ is used for the initial search along the sagittal direction. The search is conducted within the defined VOI only.
- Let the perturbed corner points of the plane defined at each step be (a, y_1, z_1) , (b, y_2, z_2) , (c, y_3, z_3) , where $a, b, c \in [s \pm \Delta_0]$. (We choose three points which define a new plane on the image boundary with $y = \text{const}$ and $z = \text{const}$. To generate a new candidate plane only x coordinate is changed because it defines the position of a plane in the volume along the sagittal direction.) Calculate the KL-measure for every plane defined by the above points.

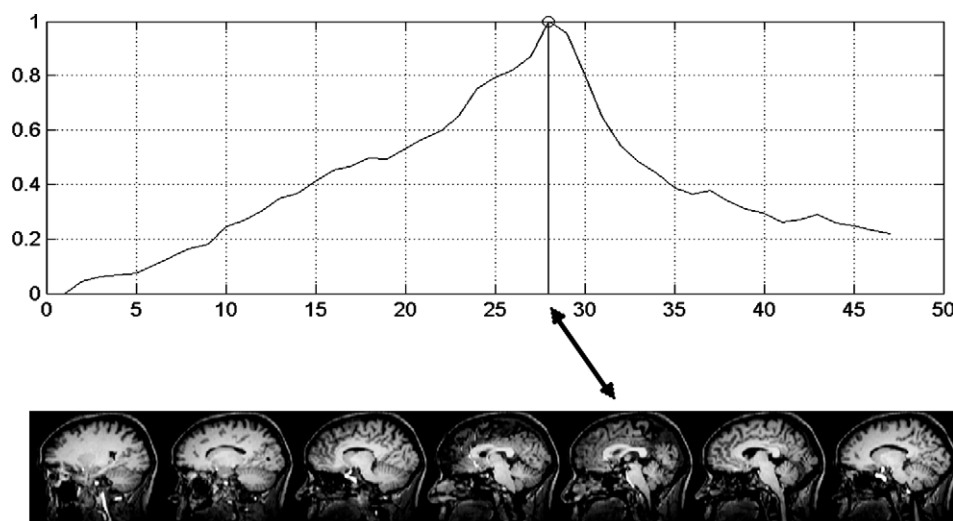


Fig. 3. Top: The values of KL-measure (normalized between 0 and 1) and the corresponding sagittal slices within the VOI (bottom). The arrow shows the correspondence between the maximum of the KL-measure and the cMSP.

- The set of corner points which gives the maximum value of the KL-measure defines the central plane for a new smaller volume of interest where the next search is performed. The new reduced step size Δ is $\lceil \Delta/2 \rceil$; the new VOI is shrunk in X direction, and has a length of 4Δ along X -direction. Calculate the KL-measure for every plane defined in the new VOI.
- Repeat the previous step while the step size is greater or equal to 1 pixel.
- The set of corner points which gives the maximum value of the KL-measure defines the corner points of the best approximation to the MSP.
- Construct the MSP from the volume data using these final set of coordinates (Fig. 4).

2.4. Data overview

A part of the data for this study was provided by Nagasaki University School of Medicine (NUSM), Japan, and Singapore General Hospital (SGH), Singapore. All the cases were anonymized. Datasets from IBSR (<http://www.cma.mgh.harvard.edu/ibsr/>) were also used to test the algorithm.

Multiple modality datasets were used in this study: MRI, MRA, and CT (Table 2). The MR datasets were acquired with various pulse sequences including T1WI (T1 weighted image), SPGR (spoiled gradient recovery) and contrast enhanced SPGR, FLAIR (fluid attenuated inversion recovery), T2WI (T2 weighted image), and PD (proton density). The datasets were not subjected to any

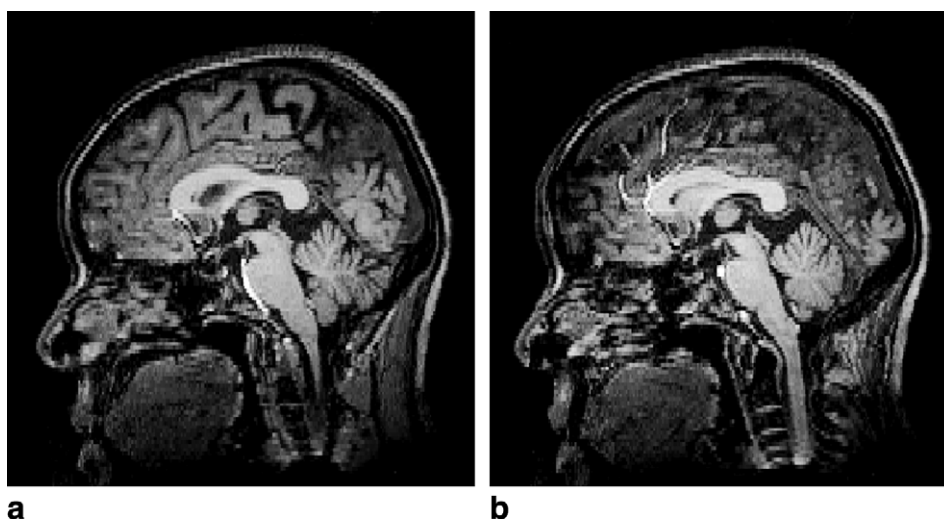


Fig. 4. MSP giving the maximum value of the KL-measure: (a) cMSP obtained after the first stage. The midline cortex is clearly visible in the plane; (b) MSP obtained after stage 2 (note a smaller amount of the midline cortex and a better delineation of the aqueduct).

Table 2
Description of datasets

Source	Modality and number of datasets	Remarks
IBSR	T1WI (20)	Ground truth was available. Datasets used for quantitative testing and validation
Clinical	T1WI (3),	Ground truth was not available. Datasets used for testing the applicability of the method
	T2WI (12),	
	SPGR (26),	
	FLAIR(12)	
	MRA (2)	
	CT (2)	
	PD (1)	Ground truth was not available. Datasets used for testing the applicability of the method
	T2WI (1)	
	FLAIR (2)	
BrainWeb	T1WI (8)	Ground truth was available. Datasets used to analyze the effects of rotation, inhomogeneity and noise

preprocessing. The slice thickness varied from 1 to 8 mm. The number of axial slices in the volumes varied from 8 to 320. Some of the datasets exhibited a high partial volume effect and severe inhomogeneity. To analyze the robustness of the method against rotation, inhomogeneity and noise the phantom data from BrainWeb (<http://www.bic.mni.mcgill.ca/brainweb/>) were used.

Datasets were chosen from the BrainWeb repository (to analyze the phantom data with different levels of noise and inhomogeneity), from the IBSR (as a standard resource utilized by researchers); and clinical brain scans were randomly chosen from our database (to show applicability for real clinical cases). We chose several cases with significant mass-effect and pathologies to demonstrate the applicability of the algorithm to abnormal brains.

2.5. Generation of the ground truth lines

For each volumetric dataset, the interhemispheric fissure ground truth lines (GTL) were drawn slice by slice (i.e., in 2D) by a neuroanatomy expert (WLN). A dedicated tool was developed to draw lines and save them. This tool enables the expert to zoom into the image and draw the lines at sub-pixel accuracy. The tool also allows the expert to propagate the ground truth line from one slice to another and to adjust it if required. The ground truth lines were drawn such that they were passing centrally through the IF.

The following quantitative measures were used to evaluate the results:

1. Angular deviation (α , in degrees) is the average of angles between the MSP and the individual ground truth lines.

2. Deviation distance (d) is the average Euclidean distance of end points of individual ground truth lines from the MSP. The deviation distance is given in pixels as the data were not reformatted during processing.
3. Deviation from planarity (α_{GTL} , in degrees) is the maximum angular deviation amongst the GTL. It is a measure of the interhemispheric fissure non-planarity.

The deviation distance depends on the marked “end points” of the GTL, so it is case-dependent.

2.6. Validation of results

We used the GTL for quantitative validation of the algorithm. If the interhemispheric fissure deviates from a plane it is unrealistic to expect a good fit of the GTL from the calculated MSP. To analyze how the “badness” of data influences the error of the MSP calculation the statistical analysis was conducted and the Kendall’s τ and the Pearson’s linear correlation tests were performed.

Some of the methods for computing the MSP are based on the least square error (LSE) criterion where the MSP is constructed as the least square error fit plane using a set of local features of the individual slices (e.g. Hu and Nowinski, 2003). A qualitative validation with respect to the LSE plane was carried out in addition to the quantitative analysis with respect to the GTL.

3. Results

Figs. 5–8 show the MSPs projected on various datasets.

Similarly to Hu and Nowinski (2003), the LSE plane was constructed using the ground truth lines, for comparison of the structures visible on the (calculated) MSP and the (ground truth-based) LSE plane (Fig. 9).

We perform non-parametric (Kendall’s τ) as well as parametric (Pearson’s linear correlation) tests for testing the hypotheses of no correlation against the alternative that there is a non-zero correlation between (i) d and α_{GTL} and (ii) α and α_{GTL} . The values obtained are summarized in Table 3.

We can conclude that the correlation between α and α_{GTL} is significantly different from zero as the corresponding probability of chance fluctuation is less than 0.05 for both the tests. However there is no significant correlation between d and α_{GTL} as the probability of chance fluctuation corresponding to Pearson’s correlation coefficient is much higher than 0.05 (we revert to the reason of this fact is in Section 4).

Results for different modalities are in Table 4.

In order to analyze the effect of noise, inhomogeneity, and slice thickness, the algorithm was tested using the BrainWeb (<http://www.bic.mni.mcgill.ca/brainweb/>) datasets (Table 5). The first digit of the dataset number stands for slice thickness, second for percentage of noise, and the last two for percentage of RF inhomogeneity (i.e., 1040 denotes the dataset has 1 mm slice thickness, 0% noise,

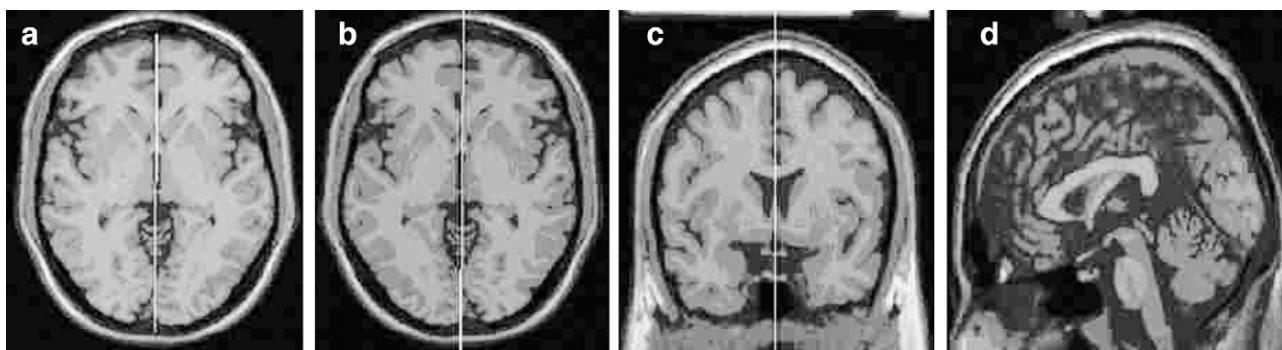


Fig. 5. Results of identification of the MSP on the phantom data with 1 mm slice thickness, noise 3% and 20% RF inhomogeneity: (a) the ground truth marked by an expert; (b) MSP projected on an axial slice; (c) MSP projected on a coronal slice; (d) the MSP.

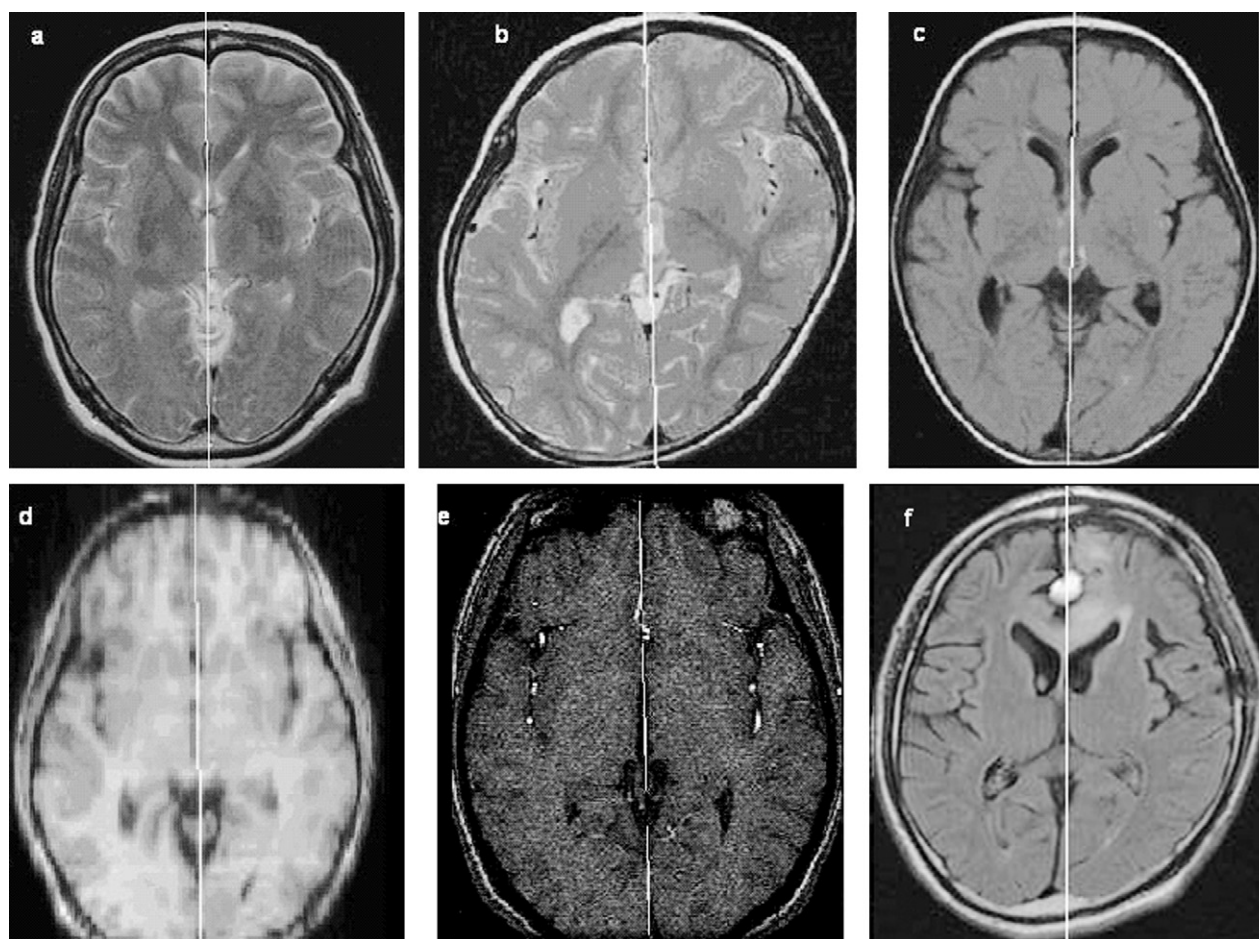


Fig. 6. Results of identification of the MSP on MR with different pulse sequences: (a) PD, (b) T2WI – distorted brain, (c) T1WI, (d) T1WI, IBSR data, (e) MRA, (f) FLAIR (tumor case).

and 40% RF inhomogeneity). Deviation from planarity α_{GTL} for the ground truth lines drawn by the expert in case of 1 mm slice thickness was 1.78° , and for 3 mm slice thickness was 1.45° .

The robustness of the algorithm for rotation was checked using the phantom data 1940 (which had slice thickness of 1 mm, 9% noise and 40% of RF inhomogeneity) with rotation in pitch, yaw and roll directions. Trilinear

interpolation was used to generate volumes after rotation. The data were rotated in both clockwise and counter clockwise from 1° to 15° in the pitch and yaw directions, from 1° to 10° in roll direction, and in combinations of these directions. The method exhibited robustness to rotation to certain degrees: (1) from -15° to 15° in each of yaw, pitch, and roll directions and (2) from -7° to 7° in roll direction combined with any other. As the results were approxi-

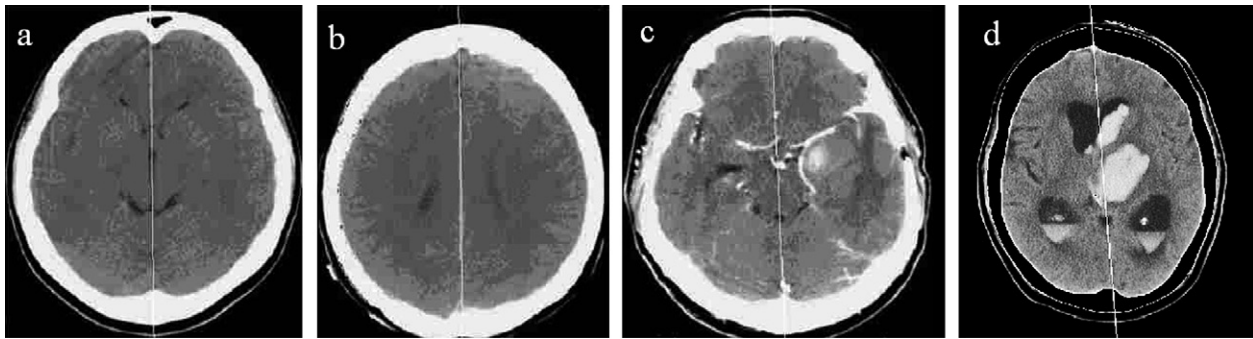


Fig. 7. Results of identification of the MSP on CT scans ($512 \times 512 \times 8$) for normal (a,b) and pathological ($256 \times 256 \times 19$) data (c,d).

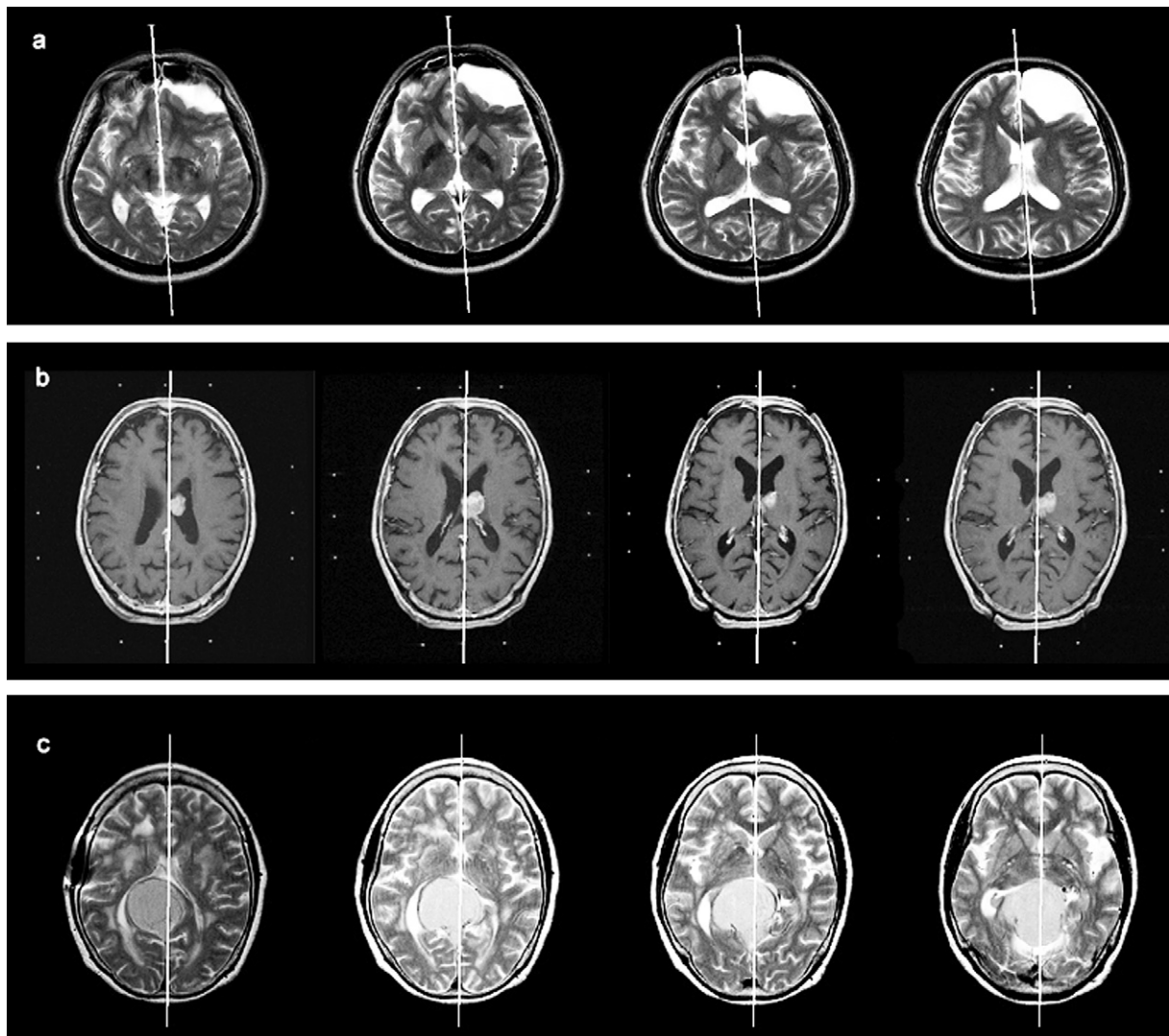


Fig. 8. Results of identification of the MSP for pathological datasets: (a) arachnoid cyst, T2WI, (b) ependymoma of the left lateral ventricle, T1WI contrast enhanced, (c) tentorial meningioma, T2WI.

mately the same for both directions, the results of counter clockwise rotation are shown in Figs. 10 and 11. Trilinear interpolation was used to construct the volume after rotation.

The time of execution of the algorithm depends on the number of voxels in the VOI. The coarse-to-fine search algorithm used has the computational complexity of $O(n)$ where n represents a number of sagittal slices in the initial

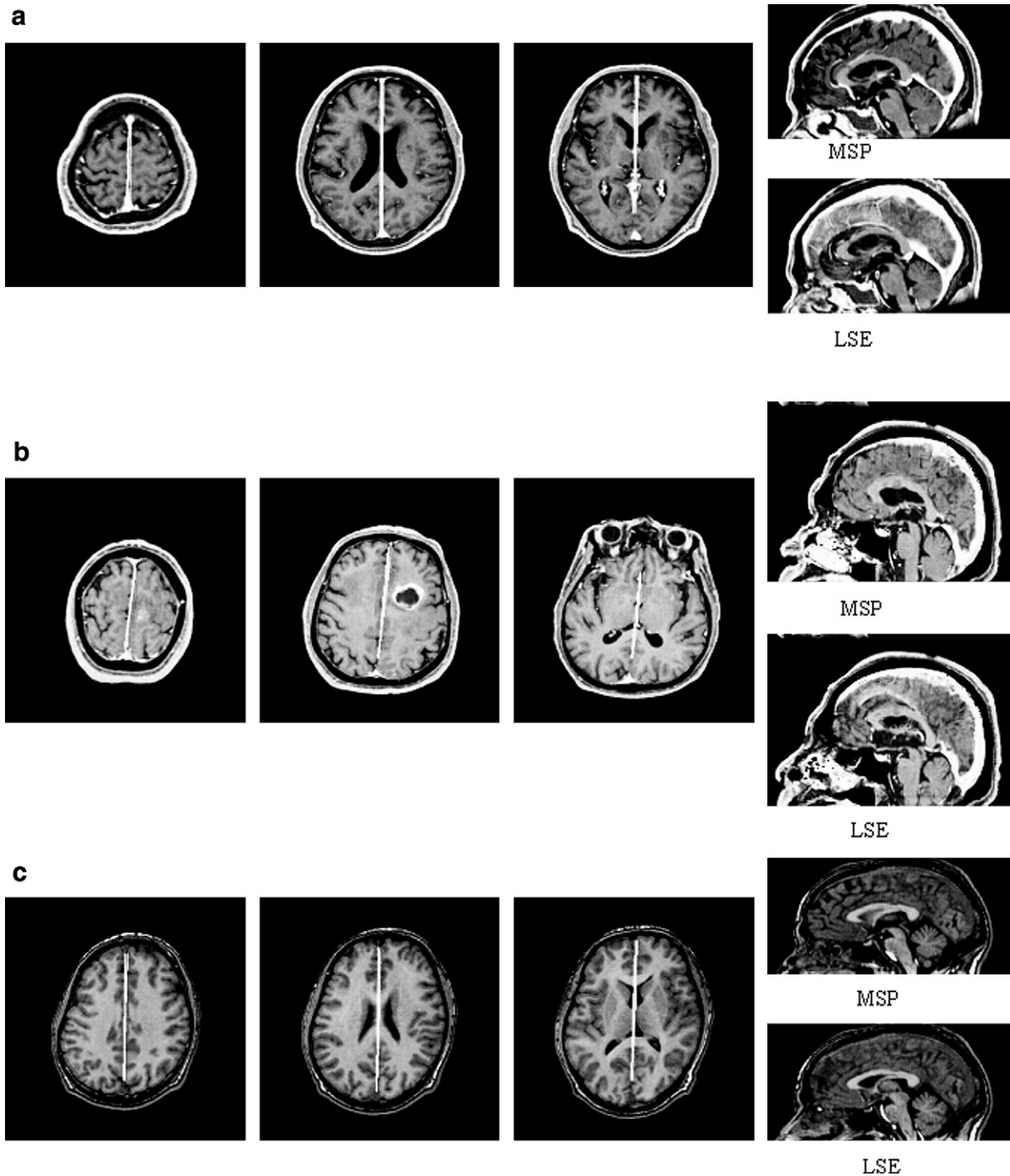


Fig. 9. Contrast enhanced SPGR axial datasets. The ground truth line is marked on each axial slice. The right most column shows the calculated MSP and the ground truth-based LSE planes for (a) normal case (note that the MSP is passing through the interhemispheric fissure as opposed to the LSE plane); (b) tumor case (note that the LSE plane deviates from the ideal MSP more than the calculated MSP; compare e.g. the pons and fourth ventricle); (c) anatomically distorted brain (note that the MSP is passing through the cerebral aqueduct as opposed to the LSE plane).

Table 3

Results for testing of the existence of a correlation between the variables

	(α, α_{GTL})	(d, α_{GTL})
Kendall's τ	0.21 (0.008)	−0.15 (0.07)
Pearson's linear correlation	0.28 (0.02)	−0.14 (0.23)

The quantities in the parenthesis indicate probability of chance fluctuation.

VOI. The algorithm, implemented in MS VC++, takes for instance less than 5 s for computation of the final MSP for the volume with dimensions $256 \times 256 \times 168$, voxel size $1.0 \times 1.1 \times 0.66$ (Windows XP, Pentium 4, 2.4 GHz, and 512 MB of RAM).

Table 4
Mean and standard deviation for different modalities

	SPGR	MRA	FLAIR	T2WI	T1WI
Number of cases	27	2	12	12	22
α_{GTL} , mean	1.47	3.29	1.05	0.72	1.52
α_{GTL} , stddev	0.67	0.37	0.67	0.55	0.68
d , mean	0.96	2.35	1.31	2.52	0.78
d , stddev	0.48	1.21	0.59	0.67	0.19
α , mean	0.79	0.15	0.44	0.56	0.62
α , stddev	0.38	0.08	0.22	0.36	0.24

Table 5
Angular deviation (α) and deviation distance (d) for the BrainWeb datasets with different noise, inhomogeneity, and slice thickness

Dataset	d	α	Dataset	d	α
1000	0.82	0.68	3000	0.58	0.64
1040	0.82	0.68	3040	0.58	0.64
1900	0.66	0.58	3900	0.58	0.64
1940	0.70	0.68	3940	0.58	0.64

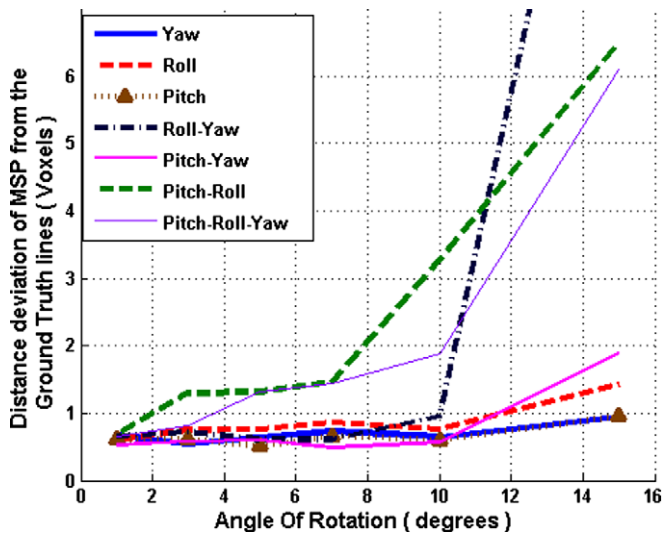


Fig. 10. Distance deviation versus pitch, roll, and yaw rotations.

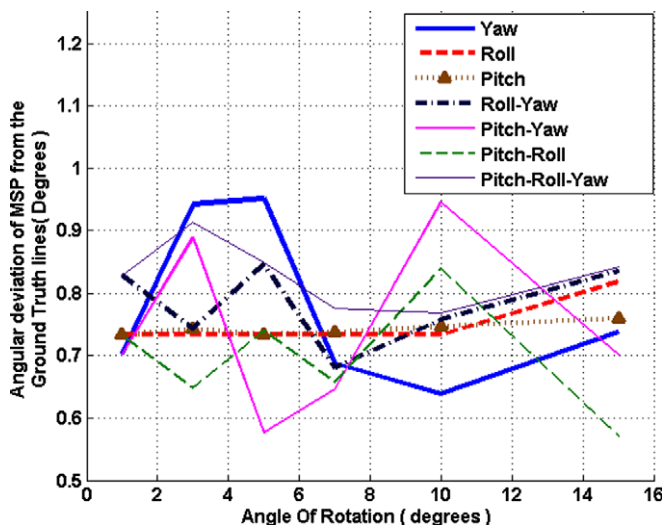


Fig. 11. Angular deviations versus pitch, roll, and yaw rotations.

4. Discussion

Brains vary from one person to another. The manifestation of brain tissues in MRI also changes from one scanner to another. There are numerous protocols, which also influence the nature of MR image. Normalization of images with respect to some standard brain image or with respect to a particular machine is difficult. In such situation, normalization of slices with respect to one reference slice and characterization of relative changes helps in intensity standardization of brain images.

4.1. Kullback–Leibler's measure and entropy

As mentioned earlier, the MSP can be defined as a plane having the minimal entropy amongst the neighboring slices. It was found empirically that entropy decreases in sagittal direction towards the MSP slice and has an inverted bell shape (Fig. 2). This is valid for ideal case (a straight IF, absence of partial volume effect, noise, RF-inhomogeneity, etc.), whereas the typical routine scans are non-ideal. These variations influence the position of the plane having the minimum entropy measure. Empirically we found that the KL-measure gives a better approximation of the cMSP and the MSP than the entropy measure. The global extremum of the entropy and the KL-measure (Fig. 2) were close to each other in most data sets studied. However in some clinical cases the initial MSP given by the entropy method did not coincide with the interhemispheric fissure.

Entropy/energy measures are based on a single plane and the objective function is not constrained. In the KL-measure based technique, the objective function is constrained by the reference plane and the solution is dependent on the reference plane. Hence the solutions obtained by energy/entropy and KL-measure based techniques are not comparable in the sense as one is unconstrained optimization whereas the other is constrained optimization. Moreover the KL-measure based technique reduces the common effects (like noise, inhomogeneity and partial volume) as there is always comparison of the studied slice with a reference slice. This property is not present in energy/entropy based methods. Hence energy and entropy methods are more sensitive to effects of noise, inhomogeneity and partial volume.

4.2. Assumptions

We assume that the roll angle is not greater than 7°.

4.3. Justification of parameters and selection of the reference slice

The parameter for selection of the reference slice at 20 mm for initial slice chosen empirically has an anatomical justification. The sagittal slice at 20 mm apart of the medial slice has large manifestation of white matter (see

e.g., Duvernoy, 1999, p. 302). It crosses the atrium and occipital horn of the lateral ventricle and additionally has CSF in the meningeal space. The medial slice of brain in the sagittal direction (Duvernoy, 1999, p. 323) contains smaller amount of WM structures and larger manifestation of CSF (in the IF). These changes are reflected in histograms (Fig. 1) and allow finding the medial sagittal slice. Different values for distance (from 5 to 40 mm) of the reference slice were analyzed and 20 mm gave the least error with respect to the ground truth. Since the value of KL-measure is considered as the global maximum at the MSP the search was restricted only for the global maximum. Further we did not come across any cases in the study where the global maximum did not coincide with the MSP.

The initial version of our algorithm had a single reference slice, which can be on any hemisphere provided the volume data is from a normal subject. If there is an affected area in the image and the reference slice goes through this area, the cMSP may be displaced and the extremum of KL-measure may not coincide with the IF because of the different intensity distribution in comparison with the normal case. To avoid the influence of the affected region on the location of cMSP, we modified the algorithm to handle the abnormal cases: instead of one reference plane, two reference planes from both hemispheres were considered. The KL-measures using each reference plane was calculated and summed up. The summed KL-measure was normalized between 0 and 1 and the position at which it has maximum was considered as the location of cMSP. The same procedure of using two reference planes was also used during the fine search process.

A step size of $\Delta_0 = \min(\lceil L/8 \rceil, 8)$ in computation of the final MSP was selected empirically for the initial search in the VOI after testing various step sizes on different datasets for faster convergence.

As the step size and the VOI decrease by half at every iteration, the search has a linear complexity.

4.4. Noise, RF-inhomogeneity, and rotation

Noise, RF-inhomogeneity, and partial volume did not affect accuracy (Table 5). This is due to the fact that both the reference slice and the MSP have more or less the same amount of noise and inhomogeneity thus canceling their effects. The algorithm was tested with 9% noise and 40% RF inhomogeneity, and there was no reduction in the accuracy of the results. In routine scans we may not come across such extreme conditions.

The distance and angular deviation errors were not greatly influenced by rotation in pitch, roll and yaw directions up to 15° (Figs. 10 and 11). It was observed that any roll rotation greater than 7° in combination with other directions lead to deviation from the ground truth lines. The distance and angular deviations could be due to the effect of trilinear interpolation and choosing the reference plane which is not parallel to the MSP. The results will

be rotation invariant if the VOI is chosen with respect to the co-ordinate system attached to the brain instead of the image coordinate system.

In comparison with the method of Liu et al. (2001) we do not correct the initial tilt of the data and the MSP is found with respect to the original orientation. Hence our algorithm works well with limited initial orientations.

4.5. Ground truth lines and MSP

Table 3 shows that the correlation between α and α_{GTL} is significantly different from zero and there is linear trend between α_{GTL} and α . The average distance and angular deviation for 75 datasets processed were 1.25 pixels and 0.63°, respectively.

GTL was marked by a neuroanatomy expert (WLN) in the linear parts of the interhemispheric fissure on axial slices. As the ground truths line segments were marked only where the IF was visible, the lengths of the ground truth line segments may vary in slices. So the distance deviation d of the MSP from the GTL does not depend purely on α_{GTL} but additionally on the length of lines from the GTL. This explains why the hypothesis of linear dependence between d and α_{GTL} cannot be accepted.

The intensity variations in GM and WM are higher than that of CSF. CSF has lesser variations in gray level, making it more homogeneous matter. The amount of CSF present in the MSP is higher than in its neighboring slices in the sagittal direction. Hence optimization performed at the second stage of algorithm aims to minimize visibility of the cortical structures at the plane, i.e., the maximum CSF representation (Fig. 4b) and the algorithm converges to the MSP.

If the original volume contains structures below the cerebellum or there is an asymmetrical IF, the algorithm finds the maximum homogeneous plane within the final VOI, often deviating from the ground truth. In such cases the error measures shows a higher deviation. The deviation was also found to be higher in the cases where the IF was wider at the anterior part of the brain than at the posterior part. In such cases, the GTL do not capture the asymmetry present in the data as the GTL are marked only in the symmetrical regions of the IF and only on axial slices in which the IF is visible. Due to this reason the data may be classified as good quality data though there are hidden asymmetries to which the algorithm will be sensitive. These are the cases which have given higher error.

4.6. MSP versus LSE

Generally the MSP is different from the LSE plane constructed from the ground truth line segments. This is illustrated in Fig. 9 where the MSP is better defined in terms of the structures visible on the MSP. Since the ground truth lines are marked based on the local features of the individual slices, the deviation of the MSP projection with respect to GTL in any slice may not be a true representative of the

error. Along with the quantitative measures, a qualitative evaluation will help in assessment of the evaluation of algorithms.

4.7. Applicability of the algorithm to different modalities, pulse sequences, and pathological datasets

Results shown in Fig. 5–8 and Table 4 indicate that the algorithm works for various modalities and pulse sequences. Figs. 7c and d and 8 show applicability of the method for pathological cases where pathology occupies several slices. The plane found by the algorithm is unaffected and passes through the IF. The applicability of the method to any neuroimages can be explained on the similar basis as applied to T1WI neuroimages.

5. Conclusion

We have applied the KL-measure to find an approximation to the MSP of volumetric morphological data. The method is simple to implement, fast in computation, robust with respect to noise and inhomogeneity effects, and does not require any preprocessing stages, like reformatting and skull stripping. The method has been tested for different pulse sequences and various modalities (MR, CT). The current results show that it is modality independent, pulse sequence independent, and handles both normal and pathological cases.

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